SOUTHWEST AI · AI FOR DATA

From Questions to Checked Answers

Use AI to answer real business questions from a spreadsheet, then check every number before you share it.

No coding | Claude + Google Sheets | All data is made up

bit.ly/sw-ai-for-data-analysis

Learn · Activity · Hands-on

Class outline

THE SIX STEPS OF ANY ANALYSIS

1 Understand the data | 2 Clean it | 3 Answer the question | 4 Chart it | 5 Read the comments | 6 Report

0–5 MIN

1 · Basics

What AI does well with data, and where it slips.

5–36 MIN

2 · Watch

All six steps on PhoneHub, a phone store, checking each answer.

36–83 MIN

3 · Do

The same six steps on your own laptop with Corner Café, a coffee shop. About half of class.

83–90 MIN

4 · Wrap up

Using AI responsibly, and what's next.

The colored strip on each slide tells you what to do: **navy** = watch and listen · **amber** = quick exercise, no tools · **teal** = work on your own laptop.

Learn Activity Hands-on

By the end, you'll be able to

1. Remove personal details from a file before you upload it
2. Turn a business question into things you can measure
3. Use AI to understand, clean and analyze a spreadsheet, and chart the answer
4. Spot AI mistakes: judge whether a result makes sense, and check it in Google Sheets
5. Sort customer comments into themes
6. Write a short report for a manager using only checked numbers

What AI does well with data, and where it slips (sometimes)

Does well

- Reads a whole spreadsheet in seconds
- Adds and averages
- Sorts comments into groups
- Writes summaries

Slips

- Trusts bad data
- Quietly fills in gaps
- Counts duplicate rows
- Makes adding mistakes
- Sounds just as confident when it’s wrong

STEP 1 · DEMO 1 · PHONEHUB

How to ask AI

A vague request like *“Analyze this data”* lets the AI pick its own question, use bad rows and skip checks. A good request has four parts:

Part	Ask yourself	In Demo 1
Role	Who would I hire to do this?	A careful data analyst
Task	What exactly do I want back? One job per request	Explain the file and suggest 5 questions it can answer
Context	What would a new coworker need to know?	What PhoneHub is, the dates, the owner’s goal
Constraints	How could this go wrong, and what rule stops it?	Row counts you can check; only questions this data can answer

> Every demo labels its prompt with these four parts. **Constraints** is where you name the mistakes to look for.

STEP 1 · DEMO 1 · PHONEHUB

Understand the data

Protect first: delete Customer_Name, Phone and Email from Reviews; fix the details hiding in R-01 and R-22.

THE PROMPT, PARTS LABELED

Role	“You are a careful data analyst.”
Task	“Explain what’s in this file and suggest 5 questions this data can answer.”
Context	“This is sales data for PhoneHub, a phone store with two shops and an online store, for January to June 2026. The owner wants to know what this data can tell him about his business.”
Constraints	“For each sheet, list the columns with an example value, what one row means, the date range and the number of rows. For each question, name the columns it needs, and only suggest questions this data can actually answer. Then list what this data can’t tell us.”

Look for: one Sales row = one sale of one model, at one store, on one date · each question names its columns · it can’t tell us *why* customers bought.

Check: =COUNTA(A:A)-1 on each tab should give 128, 13 and 40.

STEP 2 · DEMO 2 · WHY WE CLEAN FIRST

Spot the AI mistake

There’s one big mistake. **What is it, and how would you check it?**

SEPARATE EXAMPLE · MONTHLY SALES, 2025

Month	Sales ($)	Month	Sales ($)
Jan	48,200	Jul	950,000
Feb	46,900	Aug	53,900
Mar	51,300	Sep	50,700
Apr	52,800	Oct	49,800
May	54,100	Nov	52,400
Jun	55,600	Dec	58,300

WHAT THE AI SAID ABOUT 2025 SALES

“Total 2025 sales were $1,524,000, an average of $127,000 a month. July was a record month at $950,000, about 18 times a typical month. Recommendation: repeat July’s summer promotion every year.”

STEP 2 · DEMO 2 · WHAT TO LOOK FOR

Five mistakes to watch for

You’ve just caught an outlier. Here are all five, and the quick check for each.

Mistake	What it looks like	Quick check
Outlier	One value far bigger or smaller than the rest, used as if it were normal	Sort the column. Does one value tower over the others?
Missing data filled in	A month or row that isn’t in the data appears in the answer, with no note that it was estimated	Count the rows. Are they all there?
Duplicate rows	The same row counted twice, so totals are too high	Look for repeated IDs
Totals that don’t add up	A total that doesn’t match the numbers listed above it	Add the parts yourself
Wrong averages or percentages	“67% said yes” when only 4 of 12 did	Count it: how many out of how many?

STEP 2 · DEMO 2 · PHONEHUB

Clean the data

The question: is this data clean enough to trust? AI only checks what you ask for, so name the problems.

Part	What we write
Role	“You are a careful data analyst.”
Task	“Find problems in the Sales sheet. Don’t change anything yet.”
Context	“One row is one sale, January to June 2026.”
Constraints	“Check every row, not a sample. Look for blanks, duplicate Sale_IDs, brand names spelled different ways, dates outside January–June, unusual values, and rows where Net_Sales doesn’t equal Price × Quantity × (1 − Discount/100). Show a table: problem, rows, suggested fix.”

What did it miss?

Ask: “Does every model appear under only one brand?” Two Pixel 10 sales are filed under Samsung.

Fix, then check

Ask for the fixes and a before/after list: expect 126 rows. Recompute one row with `=G2*H2*(1-I2/100)`.

STEP 3 · DEMO 3 · PHONEHUB

Answer the question

First, turn each question into measures: What will he decide? Which numbers would change that decision? Which columns hold them?

Jordan asks	So we measure
Which phones should I restock?	Which brands and models sell most (units and Net Sales), compared with stock on hand from the Inventory sheet
Are online discounts too deep?	In-store vs online: the average discount and the **profit margin** (the share of each sale that's profit)

THEN: THE PROMPT ASKS FOR EXACTLY THOSE MEASURES

Role	"You are an analyst helping the store owner make a decision."
Task	"Show Net Sales by brand, the top 5 models by units, stock vs sales for each model, and in-store vs online average discount and profit margin."
Context	"Cleaned data. The owner wants to know what to restock and whether online discounts are too deep."
Constraints	"Use Net Sales = Price × Quantity × (1 − Discount/100) and profit = Net Sales − Unit cost × Quantity. Match model names between Sales and Inventory and list any that don't match. Tell me how many rows you used. Mark anything that's a guess."

STEP 3 · CHECK IT · PHONEHUB

Does the answer make sense?

A formula proves the AI added correctly, not that the answer is reasonable. Spend 30 seconds on these five questions first.

Ask	Demo 3 example
Do the parts add up to the total?	Net Sales by brand adds to $116,750.98, all of Net Sales
Were any rows lost?	Rows used = 126, the count after cleaning in Demo 2
Is every number possible?	Margins and discounts fall between 0% and 100%
Do related numbers point the same way?	Online: deeper discounts (12.5% vs 3.3%) and a thinner margin (12.4% vs 20.2%). That fits.
Does one item take an odd share?	Apple at 48% is plausible for a phone store. At 90%, look for an outlier.

> **Ask the room:** which question would have caught July’s $950,000 on slide 7?

STEP 3 · CHECK IT · PHONEHUB

Check the numbers in Google Sheets

AI can make an adding mistake and sound just as sure. Check the numbers the decision rests on: the ones going in the report.

To check	Formula, on the cleaned file	Should match
How many rows	=COUNTA(A:A)-1	126
A total for one group	=SUMIF(E:E,"Apple",K:K)	$56,293.10
Units for one group	=SUMIF(D:D,"Online",H:H)	42 units
An average for one group	=AVERAGEIF(D:D,"Online",I:I)	The AI's 12.5% online discount
One row	=G2*H2*(1-I2/100)	That row's Net_Sales
An unusual value	=MAX(K:K) vs =MEDIAN(K:K)	A max far above the median? Look

Write each checked number down. Demo 6 uses only these.

STEP 4 · DEMO 4 · PHONEHUB

Chart it

FIRST: PICK THE CHART THAT FITS THE QUESTION

You want to show	Use	PhoneHub example
How items compare	**Bar chart**, high to low	Net Sales by brand
How something changes over time	**Line chart**	Net Sales by month, January–June
How two groups compare on a measure	**Side-by-side bars**	In-store vs online: discount and margin

THEN: THE PROMPT, PARTS LABELED

Role	“You are a data analyst who makes simple, clear charts for a business owner.”
Task	“Make two charts: Net Sales by brand, highest to lowest; and in-store vs online average discount and profit margin, side by side.”
Context	“Use the cleaned data and the numbers we checked. The owner wants to see what sells and whether online discounts are too deep.”
Constraints	“Title each chart with what it shows, in plain words. Label the axes with units ($ or %). Start the bars at zero. List the numbers under each chart.”

Check: does Apple’s bar show $56,293.10? Do all bars start at zero?

STEP 5 · DEMO 5 · PHONEHUB

Read the comments

The question: what do customers complain about most? First, label reviews R-01 to R-05 yourself, so you have something to judge the AI against.

Part	What we write	Why it’s there
Role	“You are a customer-feedback analyst.”	Sets the job
Task	“Give each of the 40 reviews one category and a sentiment: positive, negative or mixed.”	**Sentiment** is the feeling behind a comment
Context	“Phone store reviews. Some are sarcastic and one is in Spanish.”	Warns the AI about the tricky ones
Constraints	“Use only these 8 categories. Counts must total 40. For the top 3 complaints, quote 2 exact comments with their review IDs.”	Fixed categories can be counted; the total catches skipped reviews; exact quotes can be checked

Check: counts total 40? One quote found word for word with Ctrl+F? How many of your 5 labels match?

STEP 6 · DEMO 6 · PHONEHUB

Report: a summary for the owner

The owner wants what the numbers mean and what to do. Here AI writes for a person, so the parts are **CRAFT**: Context, Role, Action, Format, Tone. Use only the numbers you checked.

Part	What we write	Why it’s there
Context	“Use only the numbers we checked in this chat.”	Keeps unchecked numbers out
Role	“You are an analyst reporting to the owner of PhoneHub.”	Sets the voice
Action	“Write 3 findings with numbers and 2 recommendations, each with a simple 2-week test.”	A test lets the owner try it safely
Format	“Max 150 words, bullet points.”	Short enough to be read
Tone	“Plain and friendly. Say clearly what’s a fact and what’s a guess.”	The owner isn’t a data person

YOUR TURN · CORNER CAFÉ

Meet Corner Café

You've watched every step once. Now you do all six yourself, on a business you haven't seen: a coffee shop with two cafés (Austin and Nashville) and an online store, selling beans, brewing gear, mugs and subscriptions.

THE OWNER, MAYA, ASKS

What should we promote for the holidays? Are our online discounts too deep?

Download **corner_cafe_sales.xlsx**

One Claude chat for all six steps

Stuck 2+ minutes? Raise your hand

STEP 1 · UNDERSTAND THE DATA · HANDS-ON 1

Learn what one row means

Remember Demo 1: you can’t analyze data until you know what one row means.

Protect first: delete Customer_Name, Phone and Email from Reviews. Two reviews hide personal details in their text: find and replace them.

ASK CLAUDE TO EXPLAIN THE FILE (DEMO 1 IS YOUR MODEL). ASK FOR:

One row
What one row of each sheet means

Counts
The date range and row count for each sheet

Questions
5 questions Maya could answer, and the columns each needs

Limits
What this data can’t tell us

Write down: one Sales row = _____ **Check:** =COUNTA(A:A)-1 on each tab matches the AI?

STEP 2 · CLEAN THE DATA · HANDS-ON 2

Fix what would skew the totals

Remember Demo 2: the AI only finds the problems you name.

1. Ask Claude to find problems, and name slide 8's mistakes. Use Corner Café's columns: **Order_ID**, **Category**, **Product**.
2. Hint: an **outlier**, like PhoneHub's July, is hiding in this file. Did you ask for it?
3. Ask your own "what did it miss?" question: does every product sit in only one category?
4. Ask for the fixes, a list of changed rows, and the new row count.

Check: recompute one changed row with `=G2*H2*(1-I2/100)`, using that row's number.

STEP 3 · ANSWER THE QUESTION · HANDS-ON 3

Find what to promote and restock

Remember Demo 3: start with Maya's decision, decide what to measure, then ask Claude.

STEP 1, ON PAPER: WHAT WILL MAYA DECIDE, AND WHICH NUMBERS WOULD CHANGE IT?

Maya asks	So I'll measure
What should we promote for the holidays?	
Are our online discounts too deep?	

Step 2: ask Claude for exactly those measures, with Maya's goal. Include the Net Sales and profit formulas, matching names between Sales and Inventory, rows used, and guesses marked as guesses.

Step 3, does it make sense? Categories add to the total? Same row count as after cleaning? **Step 4, check** on corner_cafe_clean.xlsx: `=SUMIF(E:E,"Whole Bean Coffee",K:K)` and `=SUMIF(D:D,"Online",H:H)`. One product name doesn't match between sheets: did you catch it?

STEP 4 · CHART IT · HANDS-ON 4

Show Maya the answer at a glance

Remember Demo 4: pick the chart that fits the question first, then ask for it.

STEP 1: WHICH CHART ANSWERS EACH QUESTION?

Maya asks	The chart I'll make
What should we promote for the holidays?	
Are our online discounts too deep?	

STEP 2: ASK CLAUDE FOR THE TWO CHARTS. YOUR REQUEST SAYS:

Who
Someone who makes simple charts for an owner

Which charts
Which two, and the order of the bars

Which data
The cleaned data, and what Maya wants to see

The rules
Titles, units, bars from zero, numbers listed

Check: Whole Bean Coffee bar = $2,858.55? About $4,730 means the outlier is back. Bars start at zero?

STEP 5 · READ THE COMMENTS · HANDS-ON 5

Find Maya’s top complaints

Remember Demo 5: label a few reviews yourself first, so you can judge the AI.

1. Label C-01 to C-05 yourself: a category, and positive, negative or mixed.
2. Ask Claude to sort all 30 reviews into the 8 categories. What must your request say so you can check the answer? Think: fixed categories, the total, exact quotes.

THE 8 CATEGORIES

- Coffee taste & freshness
- Price & value
- Staff & service
- Online ordering & shipping
- Subscription management
- Café atmosphere
- Equipment quality
- Other

Check: counts total 30 · one quote found with Ctrl+F · your labels match __ / 5

STEP 6 · REPORT · HANDS-ON 6

A checked summary for Maya

Remember Demo 6: the report uses only the numbers you checked in steps 1–5.

1. Ask Claude for a summary for Maya: 3 findings with numbers (sales and reviews) and 2 recommendations, each with a 2-week test, under 150 words, in plain language. Demo 6's prompt is your model.
2. Delete any number you didn't check.
3. Add: "Prepared with AI assistance; numbers checked by [your name]."

Check: point at each number. Can you show where you checked it? If not, check it or delete it.

Finished early? Compare Austin and Nashville, or ask Claude for a simple dashboard and check one number on it against your Google Sheets total.

Using AI responsibly

Privacy

Remove names, emails, phone numbers and ID numbers before you upload. At work, use only AI tools your organization approves.

Accuracy

You've seen today how confidently AI can be wrong, and how to catch it. Check every number before you share it.

Bias

When data or AI favors one group unfairly. A survey of online shoppers describes them, not all customers. Use AI for patterns in groups, never to judge one person.

You own it

If your name is on the report, you're responsible for its numbers, not the AI.

What you did today

Spot a mistake → Protect the data → 1 Understand the data → 2 Clean it →
3 Answer the question, and check it → 4 Chart it → 5 Read the comments → 6 Report

You watched each step once on PhoneHub, then did it yourself on Corner Café.

> **Share-out:** one finding about Corner Café, and how you checked it.

Before you leave

EXIT TICKET

One number you checked today, and one report at work you'll try this on: made-up data, or real data in an approved tool only.

TAKE-HOME

Finish any Corner Café step, make two charts, and write a one-page summary for Maya.

"Practice on made-up data, use real data only in approved tools, and check every number."